Practical No 1

Menus, Dialogs and UI Interaction

1. Develop an Android application implementing AppBar (Toolbar) and Options Menu.
2. Develop an Android application demonstrating AlertDialog and custom DialogFragment.
3. Develop an Android application implementing contextual menus and popup menus.

AIM: To study and implement UI components in Android such as AppBar, TextView, EditText, Button, ImageView, Toast, and Fragment-based UI.

THEORY:Android provides a rich set of User Interface (UI) components to build interactive applications.

Common UI Components

Component	Purpose
TextView	Displays text
EditText	Accepts user input
Button	Performs an action
ImageView	Displays images
AppBar (Toolbar)	Top app bar for titles & menus
Fragment	Reusable UI section inside activity
CheckBox / RadioButton	Selection inputs
Spinner	Dropdown menu

activity_main.xml (AppBar + Fragment Container)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <androidx.appcompat.widget.Toolbar
        android:id="@+id/toolbar"
    android:layout_width="match_parent"
```

```
    android:layout_height="?attr/actionBarSize"
    android:background="@color/purple_500"
        android:title="UI Components Demo"
        android:titleTextColor="@android:color/white"
        app:titleTextAppearance="@style/TextAppearance.AppCompat.Large"/>
    <FrameLayout
        android:id="@+id/frameContainer"
        android:layout_width="match_parent"
        android:layout_height="match_parent"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.uicomponents;
import androidx.appcompat.app.AppCompatActivity;
import androidx.fragment.app.FragmentTransaction;
import android.os.Bundle;
import androidx.appcompat.widget.Toolbar;
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Toolbar toolbar = findViewById(R.id.toolbar);
        setSupportActionBar(toolbar);
         // Load Fragment
        FragmentTransaction ft = getSupportFragmentManager().beginTransaction();
        ft.replace(R.id.frameContainer, new HomeFragment());
        ft.commit();
    }}
```

fragment_home.xml (UI Components)

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <LinearLayout
        android:orientation="vertical"
        android:padding="20dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content">
        <ImageView
            android:id="@+id/imgLogo"
            android:src="@drawable/ic_launcher_foreground"
            android:layout_width="120dp"
            android:layout_height="120dp"
            android:layout_gravity="center"/>
        <TextView
            android:id="@+id/txtTitle"
            android:text="Enter Your Name:"
            android:textSize="20sp"
            android:layout_marginTop="20dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"/>
        <EditText
            android:id="@+id/edtName"
            android:hint="Type here"
            android:padding="10dp"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>
        <Button
            android:id="@+id/btnSubmit"
```

```
        android:text="Submit"
        android:layout_marginTop="20dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
  </LinearLayout>
</ScrollView>
```

HomeFragment.java

```
package com.example.uicomponents;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;
public class HomeFragment extends Fragment {
  EditText edtName;
  Button btnSubmit;
 @Nullable
 @Override
  public View onCreateView(@NonNull LayoutInflater inflater,
   @Nullable ViewGroup container,
     @Nullable Bundle savedInstanceState) {
      View view = inflater.inflate(R.layout.fragment_home, container, false);
      edtName = view.findViewById(R.id.edtName);
      btnSubmit = view.findViewById(R.id.btnSubmit);
```

```
        btnSubmit.setOnClickListener(v -> {
            String name = edtName.getText().toString();
            Toast.makeText(getActivity(),
                    "Hello " + name,
                    Toast.LENGTH_SHORT).show();
        });
        return view;
    }}
```

